

Epic: MVP Development – Neural Edge Network

Goal: Build a working prototype demonstrating edge caching, dynamic edge functions, and SDK integration.

Ticket 1: Setup Project & Folder Structure Type: Task Priority: High Description: - Create base folder structure for MVP (`origin-server`, `edge-server`, `edge-functions`, `sdk`). - Initialize `package.json` for Node.js projects. - Set up Docker environment for local testing. Acceptance Criteria: - All folders exist with basic `package.json` and Dockerfile placeholders. Assignee: DevOps

Ticket 2: Build Origin Server Type: Task Priority: High Description: - Implement a Node.js server serving static content (`index.html`, sample images, JS). - Ensure server runs on port `3000` and serves content correctly. Acceptance Criteria: - Accessing `http://localhost:3000` returns HTML and static files. Assignee: Backend

Ticket 3: Build Edge Server (Caching Layer) Type: Task Priority: High Description: - Implement edge server using Node.js + caching directory. - Fetch missing content from origin server. - Serve cached files when available. Acceptance Criteria: - Accessing edge server returns cached content if available. - On cache miss, content is fetched from origin and stored. Assignee: Backend

Ticket 4: Build Dynamic Edge Functions Type: Task Priority: Medium Description: - Implement a basic edge function endpoint (`/api/user-profile`). - Return JSON data indicating response was served from edge. Acceptance Criteria: - Requesting `/api/user-profile` returns JSON with `cachedAtEdge: true`. Assignee: Backend

Ticket 5: Build SDK for Website Integration Type: Task Priority: Medium Description: - Implement JS SDK to fetch static content from edge server. - Implement function to call edge function endpoints for dynamic content. Acceptance Criteria: - Web page can use SDK to load static and dynamic content via edge servers. Assignee: Frontend/Integration

Ticket 6: Dockerize MVP Type: Task Priority: Medium Description: - Create `docker-compose.yml` to run origin, edge, and edge function servers locally. Acceptance Criteria: - Running `docker-compose up` starts all services successfully. Assignee: DevOps

Ticket 7: Test Local MVP Type: Task Priority: High Description: - Test edge server caching, dynamic functions, and SDK integration locally. - Measure latency improvement over origin-only requests. Acceptance Criteria: - Static content served from edge server. - Dynamic API works through edge function. - Logs show cache hits/misses correctly. Assignee: QA / Dev

Ticket 8: Documentation Type: Task Priority: Low Description: - Document folder structure, running instructions, and SDK usage. Acceptance Criteria: - README.md exists in project root with instructions to run MVP locally. Assignee: Dev / Tech Writer

Optional Future Tickets: - Predictive caching service (AI-based preloading) - Dashboard / analytics - P2P content delivery